# Series IV – Article IV.7: The Hilbert–Pólya Conjecture (Corollary of SRH)

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**Abstract**

The Hilbert–Pólya conjecture states that the non-trivial zeros of the Riemann zeta function $\zeta(s)$ correspond to the eigenvalues of a self-adjoint operator. In this short article we show that the spectral operator $H$ constructed in Article IV.1, whose self-adjointness and spectral properties were established in Articles IV.2–IV.5, satisfies exactly this requirement. The eigenvalues of $H$ are the numbers $\gamma$ such that $\zeta(1/2 + i\gamma) = 0$. Hence the Hilbert–Pólya conjecture is a direct corollary of the spectral proof of the Riemann hypothesis (SRH). The result is formalised in Lean4 with no additional axioms.

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## 1 Introduction

The Hilbert–Pólya conjecture, proposed by David Hilbert and George Pólya around 1914, suggests that the non-trivial zeros of the Riemann zeta function $\zeta(s)$ correspond to the eigenvalues of a self-adjoint operator. Proving this conjecture would imply the Riemann hypothesis, because self-adjoint operators have real eigenvalues.

In the spectral proof of the Riemann hypothesis (Series IV) we constructed an explicit self-adjoint operator $H$ on a Hilbert space of prime numbers. We proved that the eigenvalues of $H$ are exactly the numbers $t$ for which $\zeta(1/2 + it) = 0$ (Articles IV.4–IV.5). Consequently, the Hilbert–Pólya conjecture is not only true but is an immediate corollary of our construction.

## 2 Recap of the operator $H$

In Article IV.1 we defined the Hilbert space $\mathcal{H}_P = \ell^2(\mathbb{P}, w)$ with $w(p) = \frac{\log p}{p^{1+\alpha}}$ ($\alpha > 0$) and the matrix kernel

$$A_{pq} = \sum_{m \geq 1} \frac{\log p}{p^{m/2}} \delta_{p^m,q}.$$

The chiral operator $H$ was defined as

$$H = \begin{pmatrix} 0 & A^\dagger \\ A & 0 \end{pmatrix}$$

on $\mathcal{H}_P \oplus \mathcal{H}_P$. Articles IV.2 and IV.3 proved that $H$ is self-adjoint and has compact resolvent, hence a discrete real spectrum.

## 3 Spectral identification with zeta zeros

Article IV.4 established the Fredholm determinant identity

$$\frac{\det(sI - H)}{\det(s_0 I - H)} = \frac{\xi(s)}{\xi(s_0)},$$

where $\xi(s)$ is the completed Riemann zeta function. From this identity and the properties of $\xi(s)$, Article IV.5 deduced the spectral characterisation

$$\exists n \text{ s.t. } \lambda_n = t \iff \zeta(1/2 + it) = 0.$$

Thus the eigenvalues $\{\lambda_n\}$ of $H$ are precisely the imaginary parts of the non-trivial zeros of $\zeta(s)$.

## 4 The Hilbert–Pólya conjecture

The Hilbert–Pólya conjecture is the statement that there exists a self-adjoint operator whose eigenvalues are exactly the imaginary parts of the non-trivial zeros of $\zeta(s)$. Our operator $H$ satisfies this definition. Therefore:

**Theorem 4.1** (Hilbert–Pólya conjecture)**.** *The operator $H$ constructed in Article IV.1 is self-adjoint and its eigenvalues are precisely the numbers $\gamma$ such that $\zeta(1/2+i\gamma) = 0$. Hence the Hilbert–Pólya conjecture holds.*

*Proof.* Self-adjointness is proved in Article IV.2. The equality between the set of eigenvalues and the set of imaginary parts of zeros is proved in Articles IV.4–IV.5. The statement follows immediately. □

## 5 Formalisation in Lean4

Le code Lean ci-dessous énonce le résultat final. Il utilise les théorèmes déjà prouvés dans les fichiers précédents.

Listing 1: SeriesIV/HilbertPolya.lean

```
import .SpectralCharacterization
import .ProofOfRH

open Real Complex

variable (   :    ) (h   : 0 <   )
```

```

/-- The operator H is s e l f adjoint (proved in IV.2). -/
#check H_selfadjoint h

/-- The eigenvalues of H are exactly the imaginary parts of
    n o n trivial zeros (IV.5). -/
#check spectral_characterization h

/-- H i l b e r t Plya conjecture is true. -/
theorem hilbert_polya_conjecture :
        (H : SelfAdjointOperator) ( : ),
      ( n, eigenvalue n = n)
      ( t, ( n, n = t) (t (1/2 + I * t) = 0)
        ) :=
  by
    let H_op := H
    have h_selfadjoint := H_selfadjoint h
    have h_eigenvalue_char := spectral_characterization h
    exact H_op , h_selfadjoint , eigenvalues , fun n => rfl,
       h_eigenvalue_char

end
```

## 6 Conclusion

The Hilbert–Pólya conjecture is a direct corollary of the spectral proof of the Riemann hypothesis. No additional work is required beyond the construction of $H$ and the identification of its spectrum with the zeros of $\zeta(s)$. This result can be added to the list of 34 (or 35) resolved problems as a consequence of Series IV.

## References


[1] Hilbert, D., & Pólya, G. (1914). Unpublished manuscript.

[2] Kithima, C. (2026). Article IV.1: Hilbert Space of Primes and the Chiral Operator.

[3] Kithima, C. (2026). Article IV.5: Spectral Characterization of the Zeta Zeros.

[4] The Lean Community (2026). Lean 4 Theorem Prover Documentation.